

Derivation of 2-Colored Twist Formulae in Skein Theory for Odd 2-Bridge Links

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1 Introduction

The contents of this paper serve as the senior thesis for the math major, concluding a year of independent research spent under my advisor Ka Ho Wong. The first semester of such was spent studying hyperbolic knot theory, the Kauffman bracket skein module, and the Kashaev-Murakami-Murakami volume conjecture. This volume conjecture and its proof for the figure eight knot 4_1 was the main focus of study last semester, and its use of skein theory inspired the topic of study for my thesis this semester.

For a proper background of the work done last semester and how it motivated this area of study, the volume conjecture in hyperbolic knot theory relates two important knot invariants. One of these invariants is the hyperbolic volume of the knot complement $\text{Vol}(S^3 \setminus K)$, a purely geometric measure. There are very few knots which have a closed form available solution for their hyperbolic volume. The other family of invariants found in the volume conjecture are the colored Jones polynomials $J_n(K, A)$ for $n \in \mathbb{Z}_{\geq 0}$. These are known to be computable using skein-theoretic means on the knot's diagram (most notably the Kauffman bracket skein module). This is the role that skein theory plays in the story of the volume conjecture, in that it gives a method for relating hyperbolic volume and algebra on the resolving knot's diagram using the Kauffman bracket.

The colored Jones polynomials are easily defined using the Kauffman bracket as such for $n \in \mathbb{Z}_{\geq 0}$ we have:

$$J_n(K, A) = \left((-1)^n A^{n^2+2n} \right)^{-w(D)} \langle e_n \rangle_D$$

In particular, take note of the term $\langle e_n \rangle$. This is the Kauffman bracket of the knot diagram D , which is cabled using the Chebyshev polynomial e_n (defined later). Strands cabled

using this polynomial are said to be 'colored' by n , and is how we use the term 'colored' in 'colored Jones polynomial'.

Another invariant which is the normalized form of the colored Jones polynomial (with O representing the unknot) is denoted by $V_{n+1}(K, q)$:

$$V_{n+1}(K, q) = \frac{J_n(K, q^{1/4})}{J_n(O, q^{1/4})}$$

This normalized invariant is at the heart of the Murakami-Murakami-Kashaev volume conjecture, which states that for a hyperbolic knot $K \subset S^3$ (a knot whose knot complement $S^3 \setminus K$ has a hyperbolic structure), the following is true:

$$\lim_{n \rightarrow \infty} \frac{2\pi}{n} \log |V_n(K, e^{2\pi i/n})| = \text{Vol}(S^3 \setminus K)$$

In general though, it is not known whether the limit on the left side of the equation exists, and a current topic of research is relating these two invariants for more classes of hyperbolic knots. Currently, the volume conjecture is proven for only a few classes of knots, among which are the trefoil knot, figure-eight knot, and torus knots. Last semester, I explored a proof of the conjecture for the figure eight knot 4_1 . In the course of the proof, we relied on a formula from Habiro-Le for the invariant $V_n(4_1, A)$, found in 1.

$$\begin{aligned} V_n(4_1, q) &= 1 + \sum_{j=1}^{n-1} \prod_{k=1}^j (q^{(n-k)/2} - q^{-(n-k)/2}) (q^{(n+k)/2} - q^{-(n+k)/2}) \\ &= 1 + \sum_{j=1}^{n-1} \prod_{k=1}^j \{n-k\} \{n+k\} \quad \text{where } \{j\} = q^{j/2} - q^{-j/2}. \end{aligned}$$

Properties of this formula, as well as a computation for the hyperbolic volume of 4_1 allow us to equate both sides of the volume conjecture to $2V_3 = 2.02988\dots$, equal to twice the volume of an idea regular tetrahedron, proving the volume conjecture true for 4_1 .

In general, Habiro showed by studying quantum $\mathfrak{sl}_2$ invariants, for K a string knot and $n \in \mathbb{Z}_{\geq 0}$ that there is a closed form in general for $V_{n+1}(K, q)$ found in [1, Thm. 3.1].

$$V_{n+1}(K, q) = \sum_{i=0}^n a_i(K) \prod_{\substack{n+1-i \leq j \leq n+1+i \\ i \neq n+1}}^i \{j\}$$

Note that the term $a_i(K)$ does not depend on the color n . In the case of the figure eight knot, it turns out that $a_i(K) = 1$ for all $i \in \mathbb{Z}_{\geq 0}$. There are only a few classes of knots for which these terms are known, and it is why the volume conjecture has only been proven for specific cases of knots such as the trefoil knot, the figure-eight knot, twist knots, and more. While it is not a focus of this paper to explore this formula, it might be natural to ask what other properties or generalizations of $V_{n+1}(K, q)$ can be explored. In fact, a similar notion of the colored Jones polynomials exist for multiple colors (and in the two color case $J_{m,n}$ does have some relevance to the Volume conjecture for incomplete metrics).

Skein theory offers an approach for expressing closed form expressions of quantum knot invariants using the Kauffman bracket on colored knot diagrams. It is this aspect of the theory that intrigued me while I was studying knot theory last semester, and it is what I continued to explore this semester. In this paper, I will share what I have learned in my exploration of Skein theory and applications for investigating the colored Jones polynomials.

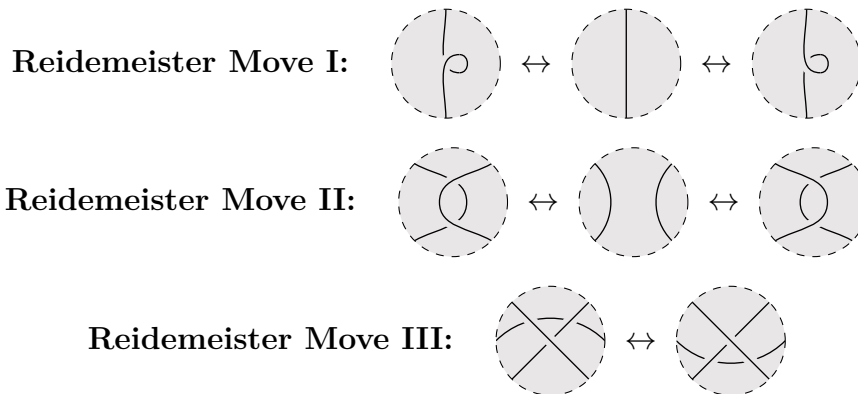
There are existing formulae I have found in the literature (2 and 3) for p full or half twists of n -colored Jones-Wenzl idempotents, but not that I had found for two differently colored strands m, n . I thought it would be interesting to generalize the formulas I had seen in these papers for colored twist formulas, and apply it to a family of links. In this paper, I have derived the formula for p full twists of (m, n) -colored strands using methods found in Yuasa's and Masbaum's papers, and applied it in the case of the odd 2-bridge link. This twist formula and its application will be shown at the end of the paper after the introductory build up through basic skein theory and the Jones-Wenzl idempotents. Skein theory's importance is seen in producing new formulae for quantum knot invariants which can prove the volume conjecture for new classes of knots and links.

I would also like to give acknowledgments, firstly to Ka Ho for being my advisor over the past year, who has served as my introduction into knot theory, and been a passionate and willing mentor. Secondly to Wutaru Yuasa, who was willing to share his workflow for creating LaTeX diagrams and whose paper as served as an inspiration for deriving generalized formulas for multiple colors.

2 Basic Knot Theory

Definition 2.1 **[[Knots and Diagrams]]:** A *knot* is a smooth embedding $K : S^1 \longrightarrow S^3$, which we often denote with K , referring the image of the map inside of S^3 . The corresponding *knot diagram* D is the projection of K to a plane (or another space when specified) with at most double intersections with crossing data. There is a similar notion for *links* which are smooth embeddings $L : \coprod S^1 \longrightarrow S^3$.

Definition 2.2 **[[Tangles and Equivalence]]:** We often talk as about *tangles* which are local diagrams of knots or links (may be thought of as smooth embeddings of arcs/circles into a diagram with the ends of each arc fixed). Knots, links, and tangles are said to be equivalent if they are *ambient isotopic*. This is described completely as having a knot/link diagram satisfying Reidemeister moves I, II, and III, as shown below as a set of tangle relations:



Links and knots which are only equivalent under Reidemeister move II and III are said to be *regularly isotopic*. Knot invariants are functions from the space of knots which preserve equivalence under ambient isotopy. The Reidemeister moves allow us to reduce this to a statement about the function being invariant under these three local transformations.

Definition 2.3 **[[q -Calculus Notation]]:** Here we will describe some of the commonly used notations for quantum integers using $k, n \in \mathbb{Z}$ and q an indeterminate. However, it should be noted that some authors prefer different variables. In this paper we will use q with $q = a^2 = A^4$.

$$\text{With } \{k\} = q^{k/2} - q^{-k/2} \text{ we define the } q\text{-bracket: } [k] = \frac{\{k\}}{\{1\}}$$

Frequently, we will find the need to use an analogue of factorials for the q bracket.

$$\{k\}! = \{k\}\{k-1\}\cdots\{1\} \text{ and } [k]! = [k][k-1]\cdots[1]$$

The q -bracket has its associated q -binomial coefficient:

$$\begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{[n]!}{[k]![n-k]!}$$

We also define the q -Pochhammer symbol and its q -binomial coefficients, which are said to be q -analogues of their original versions:

$$(q; q)_k = (q)_k := \prod_{i=1}^k (1 - q^i) \text{ and } \begin{pmatrix} n \\ k \end{pmatrix}_q = \frac{(q)_n}{(q)_k (q)_{n-k}}$$

There are multinomial forms as well:

$$\begin{pmatrix} n \\ k_1, k_2, \dots, k_m \end{pmatrix}_q = \frac{(q)_n}{(q)_{k_1} (q)_{k_2} \cdots (q)_{k_m}} \quad \left[\begin{matrix} n \\ k_1, k_2, \dots, k_m \end{matrix} \right]_q = \frac{[n]}{[k_1][k_2]\cdots[k_m]}$$

A defining motivation behind q -calculus and its functions are that when the indeterminate $q \rightarrow 1$, then the functions are equal to their classical analogues. These numbers arise in the representations of $\mathfrak{sl}_2$ with its quantum group $U_q \mathfrak{sl}_2$, but it is beyond the scope of this thesis, which considers the graphical aspects associated with skein relations and knot invariants.

3 The Kauffman Bracket Skein Module

Definition 3.1 **[[Web Spaces]]:** Denote T_{2n} the space of tangle diagrams in D^2 , the unit disk with n fixed arcs (having $2n$ distinct endpoints lying on the boundary $\partial D^2 = S^1$), and W_{2n} the $\mathbb{Q}(q^{1/4})$ -vector space with basis consisting of non-intersecting tangles diagrams in T_{2n} up to ambient isotopy. W_{2n} is called a *web space*.

Definition 3.2 **[[Kauffman Bracket]]** Then the *Kauffman Bracket* is defined as a $\mathbb{Q}(q^{1/4})$ -linear map $\langle \cdot \rangle : \mathbb{Q}(q^{1/4})T_{2n} \rightarrow W_{2n}$ with the following properties for local tangle

diagrams:

$$(1) \quad \left\langle \begin{array}{c} \text{circle with } X \end{array} \right\rangle = q^{1/4} \left\langle \begin{array}{c} \text{circle with } \cup \end{array} \right\rangle + q^{-1/4} \left\langle \begin{array}{c} \text{circle with } \cap \end{array} \right\rangle$$

$$(2) \quad \left\langle D \sqcup \begin{array}{c} \text{circle} \end{array} \right\rangle = -[2] \langle D \rangle = (-q^{1/2} - q^{-1/2}) \langle D \rangle$$

Definition 3.3: [Kauffman Bracket Skein Module]: It is also useful to imagine knots and tangles embedded in different surfaces. One particularly useful model is on the annulus $S^1 \times I$. The *Kauffman Bracket Skein Module* (KBSM) is denoted $K(S^1 \times I)$, and is the space of tangles embedded on the annulus, then reduced using the Kauffman bracket. It is isomorphic to $\mathbb{Z}[q^{\pm 1/4}][z]$ where z denotes the embedding of the circle around the hole of the annulus, and z^n is the n -cabling.

$$1 = \emptyset = \begin{array}{c} \text{circle with } \cdot \end{array} \quad z = \begin{array}{c} \text{circle with } \text{circle} \end{array} \quad z^3 = \begin{array}{c} \text{circle with } \text{circle with } \text{circle} \end{array}$$

An important set of functions arise on $K(S^1 \times I)$ as the eigenvectors of the twist operator $t : K(S^1 \times I) \longrightarrow K(S^1 \times I)$ which induces a positive "right-handed" twist on a diagram in $K(S^1 \times I)$, then resolves the induced crossings with the Kauffman bracket. The eigenvectors of the twist operator are actually a family of polynomials defined recursively:

$$e_0 = 1 \quad e_1 = z \quad e_n = ze_{n-1} - e_{n-2}$$

These have eigenvalues under the twist operator:

$$t(e_n) = \mu_n e_n \quad \text{where} \quad \mu_n = (-1)^n A^{n^2+2n} = (-1)^n q^{4n^2+8n}$$

These are called the *Chebyshev polynomials of the first kind*, and they play an important role in knot theory. A strand is said to be n -colored when it is cabled by e_n in $K(S^1 \times I)$. These polynomials are also used to define the colored Jones polynomial,

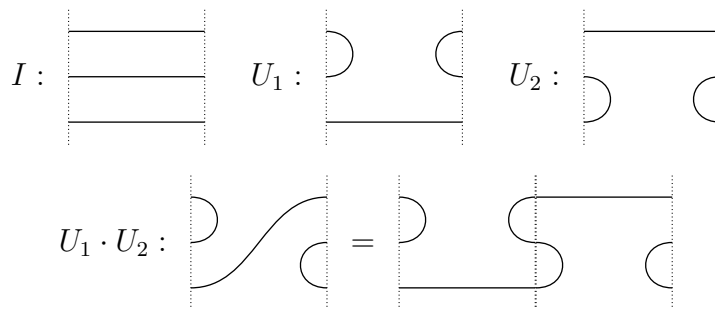
which is an invariant for any $n \in \mathbb{Z}_{\geq 0}$ and knot K with:

$$J_n(K, A) = \left((-1)^n A^{n^2+2n} \right)^{-w(D)} \langle e_n \rangle_D$$

Where $w(D)$ is the *writhe* of a knot diagram, a weighted sum of positively and negatively oriented crossings, and $\langle e_n \rangle_D$ denotes the e_n -cabling of the knot diagram D . Both the writhe and the Kauffman bracket are invariant under regular isotopy (Reidemeister moves I and II). The colored Jones polynomials are constructed to be additionally invariant under ambient isotopy by introducing the twist eigenvalue to counterbalance the coefficients induced by Reidemeister move III.

4 Temperley-Lieb Algebra & JW Idempotents

Definition 4.1 **[[Temperley-Lieb algebra]]**: An analogous way to think about colored strands is through the lens of the *Temperley-Lieb algebra* which we will denote TL_n for a diagram with n inputs and n outputs. Each input is connected to one output by an arc, and intersections are at most double crossings with crossing data. Here is a diagram showing examples of elements in TL_3 .



Composition in TL_n is as simple as regarding the outputs of the first element as the inputs of the second element, as can be seen above. Equivalence of elements in TL_n is up to ambient isotopy. In general, for non-intersecting planar graphs diagrams in TL_n , the basis elements will be the set $\{I, U_1, U_2, \dots, U_{n-1}\}$ where U_i is the element connecting the i and $i + 1$ th input as well as the i and $i + 1$ th output. I is the identity element. Every other element of TL_n is a formal sum of finite products of U_i 's and I . We will switch between the notion of Temperley-Lieb algebra and the web space W_{2n} quite often, as they are similar notions, but TL_n offers an introduction into Jones-Wenzl idempotents.

The Kauffman bracket can also be defined on TL_n in a similar sense to how it is on

W_{2n} . Closed loops evaluate again to $-[2]$. In this case, T_n is a free additive algebra over $\mathbb{Z}(q^{\pm 1/4})$ with multiplicative generators $1_n, U_1, \dots, U_{n-1}$.

Definition 4.2 [[Jones-Wenzl Idempotents]]: We define the *Jones-Wenzl idempotents* recursively as elements of TL_n satisfying the following properties:

$$f_1 = 1_n \quad f_{k+1} = f_k + \frac{[k]}{[k+1]} f_k U_k f_k$$

These elements can be shown to be the unique such elements having these properties in TL_n , and they have many useful properties such as the following:

1. **(Idempotency)**

$$f_k^2 = f_k$$

2. **(Annihilation of Turnbacks)**

$$f_k U_j = U_j f_k = 0$$

3. **(Closure Property)** Define $\text{tr}_q : TL_n \longrightarrow \mathbb{Z}[q^{\pm 1/4}]$ which takes an element of TL_n , closes each strand by connecting the i 'th input and output, then taking the Kauffman bracket of the entire diagram. Then:

$$\text{tr}_q(f_k) = (-1)^k [k+1]$$

One advantage of thinking about Jones-Wenzl idempotents in this way is to understand the decomposition and the recursive formula. However, the Jones-Wenzl idempotent can be reframed graphically in the language of W_{2n} , denoted with a white box labeled with n . It is common to refer to such strands as being colored with the label n , much in the same way as cabling with e_n . These are again defined recursively, with the following formula:

$$\left\langle \begin{array}{c} n+1 \\ \text{---} \boxed{} \text{---} \end{array} \right\rangle = \left\langle \begin{array}{c} n \\ \text{---} \boxed{} \text{---} \\ 1 \end{array} \right\rangle + \frac{[n]}{[n+1]} \left\langle \begin{array}{c} n \quad n-1 \\ \text{---} \boxed{} \text{---} \boxed{} \text{---} \\ 1 \quad 1 \end{array} \right\rangle$$

With the coloring 1 simply being the single strand. Each box labeled with n can be seen

as a 'black box' with n inputs and n outputs. These boxes are also called *clasps*. Each strand labeled with n can be imagined as containing n separate strands.

Proposition 4.3: The Jones-Wenzl idempotents also have the graphical properties:

1. $\left\langle \begin{array}{c} n \quad n \\ \text{---} \boxed{} \text{---} \end{array} \right\rangle = \text{---} \boxed{} \text{---}$
2. $\left\langle \begin{array}{c} n \\ \text{---} \boxed{} \text{---} \end{array} \begin{array}{c} k \\ \text{---} \end{array} \begin{array}{c} n-k-2 \\ \text{---} \end{array} \right\rangle = 0, \quad k \in \{0, \dots, n-2\}$
3. $\left\langle \begin{array}{c} n \\ \bigcirc \end{array} \right\rangle = (-1)^n [n+1]$

Remark 4.4: There is some conflation between cabling with e_n and coloring a strand with the Jones-Wenzl idempotent with the label n . The convention that we will take is that for a knot K colored with the Jones-Wenzl idempotent n , then:

$$\langle f_n \rangle = (-1)^{n-1} \langle e_{n-1} \rangle$$

That is, coloring a strand with the label n is the same as cabling a diagram with $(-1)^{n-1} e_{n-1}$.

Proposition 4.5: We have the following properties which help us understand how to manipulate tangle diagrams containing JW idempotents. Namely, we are able to 'split' JW clasps into smaller clasps and combine them in the same way using (1), resolve self-crossings of strands using (2), resolve self-twists using (3), and resolve partial closures using (4). For $k = 0, 1, \dots, n$, we have the following:

1. $\left\langle \begin{array}{c} n \\ \text{---} \boxed{} \text{---} \end{array} \begin{array}{c} k \\ \text{---} \end{array} \begin{array}{c} n-k \\ \text{---} \end{array} \right\rangle = \left\langle \begin{array}{c} n \quad k \\ \text{---} \boxed{} \text{---} \end{array} \begin{array}{c} n-k \\ \text{---} \end{array} \right\rangle$
2. $\left\langle \begin{array}{c} n \\ \text{---} \boxed{} \text{---} \end{array} \begin{array}{c} k \\ \text{---} \end{array} \begin{array}{c} n-k \\ \text{---} \end{array} \right\rangle = q^{k(n-k)/4} \left\langle \begin{array}{c} n \\ \text{---} \boxed{} \text{---} \end{array} \right\rangle$
3. $\left\langle \begin{array}{c} n \\ \text{---} \boxed{} \text{---} \end{array} \bigcirc^n \right\rangle = \mu_n \left\langle \begin{array}{c} n \\ \text{---} \boxed{} \text{---} \end{array} \right\rangle \quad \text{where } \mu_n = (-1)^n q^{(n^2+2n)/4}$

$$4. \left\langle \begin{array}{c} n \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ k \end{array} \right\rangle = (-1)^k \frac{[n+1]}{[n-k+1]} \left\langle \begin{array}{c} n-k \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} \right\rangle$$

For properties (2) and (3), the property holds for the oppositely oriented twist by substituting $q \rightarrow q^{-1}$.

5 Colored Skein Relations

Now that Jones-Wenzl idempotents have been defined, we are able to color any tangle diagram, and use the above properties to reduce. Our main goal remains to express the colored Jones polynomials for tangles with differently colored strands. Yuasa's paper 2 gives the following formulae and generalizations for resolving singly colored tangles in knot diagrams using the Kauffman bracket. It will be these formulae which we want to generalize.

Proposition 5.1 [Colored Kauffman Bracket Relation]: For $n \in \mathbb{Z}^+$, and two colored n -strands crossing each other in a half-twist, we can resolve them in the following way:

$$\left\langle \begin{array}{c} n \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ n \end{array} \right\rangle = \sum_{k=0}^n q^{\frac{1}{4}(-n^2+2k^2)} \binom{n}{k}_q \left\langle \begin{array}{c} n \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ n \end{array} \right\rangle$$

Proposition 5.2 [Full Twist Formula]: For $n \in \mathbb{Z}^+$, and two colored n -strands crossing each other in a full twist, we can resolve them in the following way:

$$\left\langle \begin{array}{c} n \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ n \end{array} \right\rangle = \sum_{k=0}^n (-1)^{n-k} q^{\frac{1}{2}(2k^2-n^2+k-n)} \frac{(q)_n}{(q)_k} \binom{n}{k}_q \left\langle \begin{array}{c} n \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ n \end{array} \right\rangle$$

Both of these are used extensively in the reduction of colored tangle diagrams, and a proof can be given using the above properties of Jones-Wenzl idempotents, as they are in 2. In fact, Yuasa also goes on to give a formula for repeated amounts of twists as follows:

Proposition 5.3 [p -Half Twist Formula]: Given $n \in \mathbb{Z}^+$ and two of these strands crossing in p -half twists, we can resolve them in the following way, where $k'_{i+1} = k_i - k_{i+1}$

for $i = 0, 1, \dots, p-1$.

$$\begin{aligned}
\left\langle \begin{array}{c} n \\ \text{---} \\ n \end{array} \right\rangle \begin{array}{c} p \text{ half} \\ \text{twists} \\ \dots \end{array} \begin{array}{c} n \\ \text{---} \\ n \end{array} \right\rangle &= (-1)^{pn} q^{-p(n^2+2n)} \sum_{\substack{0 \leq k_p \leq \dots \\ \leq k_1 \leq k_0 = n}} (-1)^{n-k_m} q^{(n-k_m)/2} (-1)^{\sum_{i=1}^m k_i} \\
&\times q^{\frac{1}{2} \sum_{i=1}^m (k_i^2 + k_i)} \frac{(q)_n}{(q)_{k_m}} \binom{n}{k'_1, k'_2, \dots, k'_p, k_p} \left\langle \begin{array}{c} n \\ \text{---} \\ n \end{array} \right\rangle
\end{aligned}$$

Proposition 5.4 *[[p-Full Twist Formula]]*: Similarly for $n \in \mathbb{Z}^+$ and two of these colored strands crossing in p -full twists, we can resolve them in the following way where $k'_{i+1} = k_i - k_{i+1}$

$$\begin{aligned}
\left\langle \begin{array}{c} n \\ \text{---} \\ n \end{array} \right\rangle \begin{array}{c} p \text{ full} \\ \text{twists} \\ \dots \end{array} \begin{array}{c} n \\ \text{---} \\ n \end{array} \right\rangle &= q^{-p(n^2+2n)/2} \sum_{\substack{0 \leq k_p \leq \dots \\ \leq k_1 \leq k_0 = n}} (-1)^{n-k_m} q^{(n-k_m)/2} \\
&\times q^{\sum_{i=1}^m (k_i^2 + k_i)} \frac{(q)_n}{(q)_{k_m}} \binom{n}{k'_1, k'_2, \dots, k'_p, k_p} \left\langle \begin{array}{c} n \\ \text{---} \\ n \end{array} \right\rangle
\end{aligned}$$

6 Twist Formulas for (m, n) -Colored Strands

The result at the heart of this paper is a formula for p -full twists of two differently colored strands, labeled m and n , which may be applied to link diagrams of differently colored components. For the purposes of this paper and its application to 2-bridge links in the next section, I have focused singly on the generalized colored full-twist formula. The results in the previous section can be proven by holding $m = n$. Let us build up to the proof of the p -full twist formulas of m, n -colored strands.

Proposition 6.1: $[(m, n)\text{-Colored Full Twist Formula}]$: Given $n, m \in \mathbb{Z}^+$, and assuming $m \leq n$, we have the following formula for the (m, n) -colored full twist.

$$\left\langle \begin{array}{c} \text{Diagram 1} \end{array} \right\rangle = \sum_{k=0}^m (-1)^{m-k} q^{\frac{1}{2}(2k(k+n-m)+k-m-nm)} \frac{(q)_n}{(q)_k} \binom{m}{k}_q \left\langle \begin{array}{c} \text{Diagram 2} \end{array} \right\rangle$$

Proof: This follows the proof of Lemma 4.1 in 3. Begin with the case $m = 1$, and reduce using the Kauffman bracket to recover a recursion relation. First we begin by writing:

$$\begin{aligned} \left\langle \begin{array}{c} \text{Diagram 1} \end{array} \right\rangle &= \sum_{k=0}^m C_{n,m,k} \left\langle \begin{array}{c} \text{Diagram 2} \end{array} \right\rangle \\ \left\langle \begin{array}{c} \text{Diagram 3} \end{array} \right\rangle &= q^{1/2} \left\langle \begin{array}{c} \text{Diagram 4} \end{array} \right\rangle + q^{-(n-1)/2} (1 - q^{-1}) \left\langle \begin{array}{c} \text{Diagram 5} \end{array} \right\rangle \end{aligned}$$

And it can be shown using induction on m , that:

$$\implies C_{n,m+1,k} = q^{\frac{1}{2}(n-2k)} C_{n,m,k} + q^{\frac{1}{2}(n-2k+1)} (1 - q^{-n+k-1}) C_{n,m,k-1}$$

for which

$$C_{n,m,k} = q^{\frac{1}{2}m(n-k)} \left[\begin{array}{c} m \\ k \end{array} \right]_q \prod_{j=n-k+1}^n (1 - q^{-j})$$

the unique solution is provided by Masbaum in 3. This solution must be unique due to the 'binomial' like recurrence property above. That is, that $C_{n,0,0} = 1$ and $C_{n,m,k} = 0$ for $k < 0$ or $k > m$, and also notice that the product only needs to go up to m . We can rewrite $C_{n,m,k}$ to better fit our structure, using the following properties:

$$\left[\begin{array}{c} m \\ k \end{array} \right]_q = q^{-(m-k)(k)/2} \binom{m}{k}_q \quad (q')_n = (-1)^n q^{-n(n+1)/2} (q)_n$$

We will also make the substitutions using the index $k \longrightarrow m-k$, $m \longrightarrow k$, $n \longrightarrow k+n-m$,

in order to have the diagram fit the form as stated above.

$$\begin{aligned}
&\Rightarrow C_{n,m,k} = q^{m(k+n-m)/2} \left[\begin{matrix} m \\ m-k \end{matrix} \right]_q \cdot \frac{(q')_n}{(q')_{k+n-m}} \\
C_{n,m,k} &= q^{m(k+n-m)/2} \cdot q^{-(m-k)k/2} \cdot \left(\begin{matrix} m \\ m-k \end{matrix} \right)_q \cdot \frac{(-1)^n q^{-n(n+1)/2}}{(-1)^{k+n-m} q^{-(k+n-m)(k+n-m+1)/2}} \cdot \frac{(q)_n}{(q)_{k+n-m}} \\
&\Rightarrow \left\langle \begin{array}{c} \text{Diagram 1} \end{array} \right\rangle = \sum_{k=0}^m (-1)^{m-k} q^{\frac{1}{2}(2k(k+n-m)+k-m-nm)} \frac{(q)_n}{(q)_k} \left(\begin{matrix} m \\ k \end{matrix} \right)_q \left\langle \begin{array}{c} \text{Diagram 2} \end{array} \right\rangle
\end{aligned}$$

Proposition 6.2 [Sliding Formula]: For the colored full twist, this formula allows us to slide over components, and ends up being instrumental in the proof of the $p(m, n)$ -colored twist formulas.

$$\left\langle \begin{array}{c} \text{Diagram 1} \end{array} \right\rangle = \frac{\mu_{m-k} \mu_{n-k}}{\mu_m \mu_n} \left\langle \begin{array}{c} \text{Diagram 2} \end{array} \right\rangle$$

Proof: We're able to slide the rightmost JW idempotents to the right side, and reduce using the properties in Proposition 4.5.

$$\begin{aligned}
\left\langle \begin{array}{c} \text{Diagram 1} \end{array} \right\rangle &= \left\langle \begin{array}{c} \text{Diagram 2} \end{array} \right\rangle = \mu_k^{-1} q^{n(n-k)/4} q^{m(m-k)/4} \left\langle \begin{array}{c} \text{Diagram 3} \end{array} \right\rangle \\
&= \mu_k^{-2} q^{-n(n-k)/2} q^{-m(m-k)/2} \left\langle \begin{array}{c} \text{Diagram 4} \end{array} \right\rangle = \frac{\mu_{m-k} \mu_{n-k}}{\mu_m \mu_n} \left\langle \begin{array}{c} \text{Diagram 5} \end{array} \right\rangle
\end{aligned}$$

Note that $q^{-n(n-k)/2} = \mu_k \mu_{n-k} \mu_n^{-1}$ and similarly, $q^{-m(m-k)/2} = \mu_k \mu_{m-k} \mu_m^{-1}$ is where the above substitution came from.

Corollary:

$$\left\langle \begin{array}{c} \text{Diagram 1} \end{array} \right\rangle = \left(\frac{\mu_{m-k} \mu_{n-k}}{\mu_m \mu_n} \right)^{1/2} \left\langle \begin{array}{c} \text{Diagram 2} \end{array} \right\rangle$$

Theorem 6.3 $\llbracket p$ Full Twists of (m, n) -Colored Strands. $\rrbracket$: For $m, n \in \mathbb{Z}^+$, and assuming $m \leq n$, and for $k'_{i+1} = k_i - k_{i+1}$, we have the following formula for p -full twists of (m, n) -colored strands:

$$\left\langle \begin{array}{c} n \\ \text{---} \end{array} \text{---} \text{---} \text{---} \begin{array}{c} n \\ \text{---} \end{array} \right\rangle = q^{\frac{-p}{2}(mn+2m)} \sum_{0 \leq k_p \leq \dots \leq k_1 \leq k_0 = m} (-1)^{m-k_p} q^{\frac{m-k_p}{2}} \\ \times q^{\sum_{i=1}^p [k_i^2 + k_i(n-m+1)]} \frac{(q)_n}{(q)_{k_p+n-m}} \binom{m}{k'_1, \dots, k'_p, k_p}_q \left\langle \begin{array}{c} n \\ \text{---} \end{array} \text{---} \text{---} \text{---} \begin{array}{c} n \\ \text{---} \end{array} \right\rangle$$

$$\langle \sigma_{m,n}^{p-i}(k_i) \rangle = \left\langle \begin{array}{c} \text{\textit{n}} \quad \text{\textit{k}}_i \\ \text{\textit{m}} - \text{\textit{k}}_i \quad \text{\textit{k}}_i + \text{\textit{n}} - \text{\textit{m}} \\ \text{\textit{m}} \end{array} \begin{array}{c} p-i \text{ full} \\ \text{twists} \\ \dots \end{array} \begin{array}{c} \text{\textit{k}}_i \quad \text{\textit{n}} \\ \text{\textit{k}}_i + \text{\textit{n}} - \text{\textit{m}} \quad \text{\textit{m}} - \text{\textit{k}}_i \\ \text{\textit{k}}_i + \text{\textit{n}} - \text{\textit{m}} \quad \text{\textit{m}} \end{array} \right\rangle$$

Notice that the diagram we are trying to reduce in the above theorem is $\langle \sigma_{m,n}^p(k_0) \rangle$ where $k_0 = m$. Defining it this way allows us to build our result up inductively. Using the sliding formula of proposition 6.2 as well as the single (m,n) -colored full twist reduction formula of 6.1, we are able to reduce a full twist at the rightmost end of the diagram, then 'slide' its vestigial components completely to the right by applying the sliding formula $p - i - 1$ times in the following way:

$$\langle \sigma_{m,n}^{p-i}(k_i) \rangle = \sum_{k_{i+1}=0}^{k_i} q^{\frac{1}{2}(2k_{i+1}^2 + 2k_{i+1}(n-m) - k_i(k_i + n-m) - k_i + k_{i+1})} (-1)^{k_i - k_{i+1}} \frac{(q)_{k_i + n-m}}{(q)_{k_{i+1} + n-m}} \binom{k_i}{k_i - k_{i+1}}_q$$

$$\begin{aligned}
&= \sum_{k_{i+1}=0}^{k_i} q^{\frac{1}{2}(2k_{i+1}^2 + 2k_{i+1}(n-m) - k_i(k_i+n-m) - k_i+k_{i+1})} (-1)^{k_i-k_{i+1}} \frac{(q)_{k_i+n-m}}{(q)_{k_{i+1}+n-m}} \binom{k_i}{k_i-k_{i+1}}_q \\
&\quad \times \left(\frac{\mu_{k_{i+1}} \mu_{k_{i+1}+n-m}}{\mu_{k_i} \mu_{k_i+n-m}} \right)^{p-i-1} \langle \sigma_{m,n}^{(p-i-1)}(k_{i+1}) \rangle
\end{aligned}$$

Then, we have a recurrence relation for $\langle \sigma_{m,n}^{(p-i)}(k_i) \rangle$ in terms of $\langle \sigma_{m,n}^{(p-i-1)}(k_{i+1}) \rangle$, which allows us to express $\langle \sigma_{m,n}^{(p)}(k_0) \rangle$ in terms of $\langle \sigma_{m,n}^{(0)}(k_p) \rangle$ where $k_0 = m$, with coefficients as a sum of indices of k_i for $i = 0, 1, \dots, p-1$. Let us use the label $k'_{i+1} = k_i - k_{i+1}$. Many of the terms end up telescoping and simplifying:

$$\begin{aligned}
\langle \sigma_{m,n}^{(p)}(k_0) \rangle &= \sum_{\substack{0 \leq k_p \leq \dots \\ \leq k_1 \leq k_0=0}} (-1)^{m-k_p} \cdot \left(\prod_{i=0}^{p-1} q^{\frac{1}{2}(2k_{i+1}^2 - k_i^2)} q^{\frac{(n-m)}{2}(2k_{i+1} - k_i)} q^{k_{i+1} - k_i} \right) \cdot \frac{(q)_n}{(q)_{k_p+n-m}} \\
&\quad \times \binom{m}{k'_1, k'_2, \dots, k'_p, k_p} \times q^{\frac{1}{2}(p-1)(-2m-mn)} q^{\frac{1}{2} \sum_{i=1}^{p-1} [k_i^2 + k_i(n-m+2)]} \times \langle \sigma_{m,n}^0(k_p) \rangle
\end{aligned}$$

We can simplify the algebra by considering the total exponential E from the q term and F from the μ term, then verifying this matches the $m = n$ case of Yuasa.

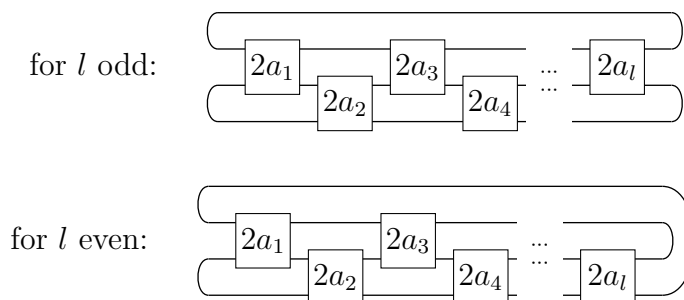
$$\begin{aligned}
E &= \frac{1}{2} \left(-k_0^2 + 2k_p^2 + \sum_{i=1}^{p-1} k_i^2 \right) + \frac{(n-m)}{2} \left(-k_0 + 2k_p + \sum_{i=1}^{p-1} k_i \right) + \frac{-k_0 + k_p}{2} \\
F &= \frac{1}{2} \left((p-1)(-2m-mn) + \sum_{i=1}^{p-1} [k_i^2 + k_i(n-m+2)] \right) \\
E + F &= \frac{1}{2} \left(-m^2 + 2k_p^2 + (n-m)(2k_p - m) - m + k_p + (p-1)(-2m-mn) \right) \\
&\quad + \frac{1}{2} \sum_{i=1}^{p-1} [2k_i^2 + 2k_i(n-m+1)] \\
\implies E + F &= \frac{1}{2} \left(2k_p(k_p n - m) + 2k_p^2 + m + k_p + p(-2m-mn) \right) + \sum_{i=1}^{p-1} k_i^2 + k_i(n-m+1) \\
&= \frac{1}{2} \left(m - k_p + p(-2m-mn) \right) + \sum_{i=1}^p [k_i^2 + k_i(n-m+1)]
\end{aligned}$$

Which gives the exponent we were seeking in the above formula. One can see that setting

$m = n$ gives us the formula found in Proposition 5.4.

7 Application

Definition 7.1 **[[2-Bridge Links]]**: One useful place to apply our theory for full twists of differently colored strands is in the 2-colored Jones polynomials of a class of knots known as *2-bridge links*. These are composed of series of full twists, labelled $L = [2a_1, 2a_2, \dots, 2a_l]$ with $a_1, \dots, a_l \in \mathbb{Z}_{\neq 0}$. Negative a_i correspond to negatively oriented twists and are associated with a swapping in the full-twist formula of $q \rightarrow q^{-1}$.



Yuasa explored $\mathfrak{sl}_2$ and $\mathfrak{sl}_3$ colored Jones polynomials for 2 bridge links of the same color in 2. In this paper I wanted to generalize the formula for differently colored knots. Note, that in the case of l being even, the diagram will actually result in being a one-colored knot. For the purposes of using our generalized formula, we will focus on the case where l is odd, and the two components separate and allow us to color them with m and n . Let us begin to generalize using the same proof method of Yuasa to find the 2-colored Jones polynomial for the odd 2-bridge links/

Theorem 7.2 **[[Two-Color Jones Invariant]]**: The colored Jones polynomials are usually defined in terms of one color, but there are analogues which take into account multiple strands of different colors. In our case, two will suffice. Let $L_{m,n} = L_m \cup L_n$, the two components of a link which are colored by $m, n \in \mathbb{Z}^+$ respectively. Then the *two-color Jones polynomials* are given by:

$$J_{m,n}(L_{m,n}, A) = \mu_m^{-w(L_m)} \mu_n^{-w(L_n)} \langle L_{m,n} \rangle$$

Lemma 7.3

$$\left\langle \begin{array}{c} \text{Diagram: A braid with two strands, top labeled } n \text{ and bottom labeled } m. \text{ There are } k \text{ crossings between them.} \end{array} \right\rangle = (-1)^{n+m-k} \frac{[n+1][m+1]}{[k+1]}$$

This directly follows from the properties in Proposition 4.4 and 4.5, and helps in the computation of the colored Jones polynomials of the odd 2-bridge links.

Notation: For the sake of cleaning up the formulas used in the next theorem. Let us define $K_0 = n$ and $K_j = m - k_{|a_j|}^{(j)}$, with $k_0^{(j)} = K_j$ and $k_{i+1}^{(j+1)'} = k_i^{(j)} - k_{i+1}^{(j)}$.

$$\text{Twist Sign: } \epsilon_{j+1} = \frac{a_{j+1}}{|a_{j+1}|}$$

$$\text{Sliding Formula: } \delta_{n,m}(K_j) = \left(\frac{\mu_{K_j} \mu_{K_j+n-m}}{\mu_n \mu_m} \right)^{\epsilon_{j+1}}$$

$$\begin{aligned} \text{Full Twist Coefficient: } \gamma_{m,n}(k_1^{(j+1)}, \dots, k_{|a_{j+1}|}^{(j+1)}; q) &= (-1)^{m-k_{|a_{j+1}|}^{(j+1)}} q^{-\frac{|a_{j+1}|}{2}(mn+2m)} q^{\frac{1}{2}(m-k_{|a_{j+1}|}^{(j+1)})} \\ &\times q^{\sum_{i=1}^{|a_{j+1}|} (k_i^{(j+1)})^2 + (k_i^{(j+1)})(n-m+1)} \frac{(q)_n}{(q)_{k_{|a_{j+1}|}^{(j+1)} + n-m}} \binom{m}{k_1^{(j+1)'}, k_2^{(j+1)'}, \dots, k_{|a_{j+1}|}^{(j+1)'}, k_{|a_{j+1}|}^{(j+1)}} \end{aligned}$$

Theorem 7.4 [2-Colored Jones Polynomial of Odd (m, n) -Colored Two-Bridge Links]

For $m, n \in \mathbb{Z}^+$, $m \leq n$ and $L = [2a_1, \dots, 2a_l]$ with l odd, let us define $L_{n,m}$ as the coloring of L with the top strand replaced by n and the bottom strand replaced by m as such:

$$L_{(n,m)} := \begin{array}{c} \text{Diagram: A braid with two strands, top labeled } n \text{ and bottom labeled } m. \text{ There are } l \text{ crossings, each labeled } 2a_i. \end{array} \quad \text{with } l \text{ odd}$$

Then:

$$\begin{aligned} J_{m,n}(K, q) &= \left(\prod_{\substack{0 \leq j \leq l-1 \\ j \text{ even}}} \sum_{\substack{0 \leq k_{|a_{j+1}|}^{(j+1)} \leq \dots \leq k_1^{(j+1)} \leq K_j}} \delta_{K_j, K_j+n-m}(m - K_j) \gamma_{K_j, K_j+n-m} \left(k_1^{(j+1)}, \dots, k_{|a_{j+1}|}^{(j+1)}; q^{\epsilon_{j+1}} \right) \right) \\ &\times \left(\prod_{\substack{0 \leq j \leq l-1 \\ j \text{ odd}}} \sum_{\substack{0 \leq k_{|a_{j+1}|}^{(j+1)} \leq \dots \leq k_1^{(j+1)} \leq K_j}} \delta_{m,m}(m - K_j) \gamma_{K_j, K_j} \left(k_1^{(j+1)}, \dots, k_{|a_{j+1}|}^{(j+1)}; q^{\epsilon_{j+1}} \right) \right) \end{aligned}$$

$$\times \mu_m^{-2(a_2+a_4+\dots+a_{l-1})} \mu_n^{-2(a_1+a_3+\dots+a_l)} (-1)^{n+K_l} \frac{[n+1][m+1]}{[m-K_l+1]}$$

Proof: We have already defined the full twist formula for m, n knots as well as the sliding formula, so we are able to reduce everything individually using the colored Kauffman bracket. Let us reduce from left to right, calculating the Kauffman bracket of the following diagrams (from which the two bridge link is composed)

Reduction for middle twists:

$$\begin{aligned} & \left\langle m - K_j \left(\begin{array}{c} \text{Diagram 1: A box labeled } 2a_{j+1} \text{ with } n-m+K_j \text{ strands on the left, } m-K_j \text{ strands on the right, and } K_j \text{ strands at the bottom.} \end{array} \right) \right\rangle \\ &= \delta_{n,m} (m - K_j)^{2|a_{j+1}|} \left\langle m - K_j \left(\begin{array}{c} \text{Diagram 2: A box labeled } 2a_{j+1} \text{ with } n-m+K_j \text{ strands on the left, } m-K_j \text{ strands on the right, and } K_j \text{ strands at the bottom.} \end{array} \right) \right\rangle \\ &= \delta_{n,m} (m - K_j)^{2|a_{j+1}|} \sum_{0 \leq k_{|a_{j+1}|}^{(j+1)} \leq \dots \leq k_1^{(j+1)} \leq K_j} \gamma_{K_j, K_j+n-m}(k_1^{(j+1)}, \dots, k_{|a_{j+1}|}^{(j+1)}; q^{\epsilon_{j+1}}) \\ & \quad \times \left\langle m - k_{|a_{j+1}|}^{(j+1)} \left(\begin{array}{c} \text{Diagram 3: A box labeled } 2a_{j+1} \text{ with } n-m+k_{|a_{j+1}|}^{(j+1)} \text{ strands on the left, } m-k_{|a_{j+1}|}^{(j+1)} \text{ strands on the right, and } k_{|a_{j+1}|}^{(j+1)} \text{ strands at the bottom.} \end{array} \right) \right\rangle \\ &= \delta_{n,m} (m - K_j)^{2|a_{j+1}|} \sum_{0 \leq k_{|a_{j+1}|}^{(j+1)} \leq \dots \leq k_1^{(j+1)} \leq K_j} \gamma_{K_j, K_j+n-m}(k_1^{(j+1)}, \dots, k_{|a_{j+1}|}^{(j+1)}; q^{\epsilon_{j+1}}) \\ & \quad \times \left\langle K_{j+1} \left(\begin{array}{c} \text{Diagram 4: A box labeled } 2a_{j+1} \text{ with } n-K_{j+1} \text{ strands on the left, } K_{j+1} \text{ strands on the right, and } m-K_j \text{ strands at the bottom.} \end{array} \right) \right\rangle \end{aligned}$$

Reduction for lower twists:

$$\left\langle K_j \left(\begin{array}{c} \text{Diagram 5: A box labeled } 2a_{j+1} \text{ with } n-K_j \text{ strands on the left, } K_j \text{ strands on the right, and } m-K_j \text{ strands at the bottom.} \end{array} \right) \right\rangle$$

$$\begin{aligned}
&= \delta_{m,m}(m - K_j)^{2|a_{j+1}|} \left\langle K_j \left(\begin{array}{c} \text{Diagram with } n \text{ strands, } n - K_j \text{ and } K_j \text{ crossings, and a box labeled } 2a_{j+1} \end{array} \right) \right\rangle \\
&= \delta_{m,m}(m - K_j)^{2|a_{j+1}|} \sum_{0 \leq k_{|a_{j+1}|}^{(j+1)} \leq \dots \leq k_1^{(j+1)} \leq K_j} \gamma_{K_j, K_j}(k_1^{(j+1)}, \dots, k_{|a_{j+1}|}^{(j+1)}; q^{\epsilon_{j+1}}) \\
&\quad \times \left\langle K_j \left(\begin{array}{c} \text{Diagram with } n \text{ strands, } n - K_j \text{ and } K_j \text{ crossings, and } k_{|a_{j+1}|}^{(j+1)} \text{ strands} \end{array} \right) \right\rangle \\
&= \delta_{m,m}(m - K_j)^{2|a_{j+1}|} \sum_{0 \leq k_{|a_{j+1}|}^{(j+1)} \leq \dots \leq k_1^{(j+1)} \leq K_j} \gamma_{K_j, K_j}(k_1^{(j+1)}, \dots, k_{|a_{j+1}|}^{(j+1)}; q^{\epsilon_{j+1}}) \\
&\quad \times \left\langle m - K_{j+1} \left(\begin{array}{c} \text{Diagram with } n - m + K_{j+1} \text{ and } m - K_{j+1} \text{ crossings} \end{array} \right) \right\rangle
\end{aligned}$$

Then in computing the bracket of the entire link, we should handle a_j 's separately for j odd versus j even:

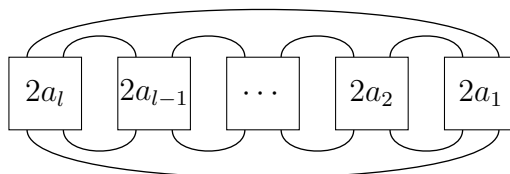
$$\begin{aligned}
\langle L_{n,m} \rangle &= \left(\prod_{\substack{0 \leq j \leq l-1 \\ j \text{ even}}} \sum_{0 \leq k_{|a_{j+1}|}^{(j+1)} \leq \dots \leq k_1^{(j+1)} \leq K_j} \delta_{n,m}(m - K_j) \gamma_{K_j, K_j+n-m}(k_1^{(j+1)}, \dots, k_{|a_{j+1}|}^{(j+1)}; q^{\epsilon_{j+1}}) \right) \\
&\quad \times \left(\prod_{\substack{0 \leq j \leq l-1 \\ j \text{ odd}}} \sum_{0 \leq k_{|a_{j+1}|}^{(j+1)} \leq \dots \leq k_1^{(j+1)} \leq K_j} \delta_{m,m}(m - K_j) \gamma_{K_j, K_j}(k_1^{(j+1)}, \dots, k_{|a_{j+1}|}^{(j+1)}; q^{\epsilon_{j+1}}) \right) \\
&\quad \times \left\langle \begin{array}{c} \text{Diagram with } n \text{ strands, } m - K_l \text{ crossings, and } m \text{ strands} \end{array} \right\rangle
\end{aligned}$$

For which the final diagram resolves to $(-1)^{n+K_l} \frac{[n+1][m+1]}{[m-K_l+1]}$ by the closure property of Lemma 7.2.

This gives us the 2-colored Jones polynomial for this case of odd (m, n) -colored 2-bridge links. For the case of $m = n$, Yuasa recovers a formula for the colored Jones

polynomial from this formula. Another formula may result by switching the n and m colors on the link components, but I argue that the formula can be computed in the same way as we have done above.

Besides just the odd 2-bridge links, there are more classes of knots which I think would be interesting to reduce using the generalized full-twist formula. Given more time to explore I would have also liked to explore formula for braids of colored Jones-Wenzl idempotents, as well as the formula for the half twist and where it can be applied for classes of knots. One class of knots that I think would be worth exploring is the class of Pretzel links for which we could find formulae for arbitrarily many differently colored components:



8 Bibliography

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